The Journal of Infectious Diseases
MAJOR ARTICLE

Rotavirus-Specific IgA and IgG Patterns During the First 2 Years of Life in the PREVAIL Birth Cohort

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Background. Mitigating the underlying causes of differential immunity from rotavirus infection and vaccination requires a comprehensive understanding of rotavirus natural history and immune response. A US-based birth cohort provides a unique opportunity to longitudinally examine rotavirus immune response.

Methods. Serum was collected from children at birth (cord blood), 6 weeks, and 6, 12, 18, and 24 months of age. Maternally-derived rotavirus-specific IgG (birth samples) and rotavirus-specific IgA and IgG (6-week through 24-month samples) were measured via enzyme immunoassay. Stool samples were collected weekly and during episodes of acute gastroenteritis to identify rotavirus infections. Serum samples prior to and after infection or vaccination were compared to assess their impact on IgA geometric mean titers (GMT). A generalized estimating equation was fit to identify predictors of IgA GMT.

Results. The median change in IgA GMT from vaccination comparing 6-week and 6-month samples was 38.4 U/mL (IQR 4.4–138.1). The change in IgA following infection varied widely (median = 64.5 [IQR 2.0–192.9]) with preinfection and postinfection GMTs of 35.7 U/mL (95% CI: 21.4–59.6) and 102.6 U/mL (95% CI: 41.1–256.0), respectively. In adjusted analyses, IgA GMT was higher when mother and/or child was a secretor (ratio of means [RoM] = 1.3, 95% CI: 1.1–1.4) compared with nonsecretor mother–child pairs. Being fully vaccinated was more strongly associated with IgA GMT in the first (RoM = 1.9, 95% CI: 1.7–2.0) compared with the second year of life (RoM = 1.3, 95% CI: 1.0–1.7).

Conclusions. This longitudinal assessment of rotavirus-specific immune response contributes to our understanding of multifactorial drivers of rotavirus immunity in a postvaccine setting.

Keywords. rotavirus; antibodies; vaccines; child; birth cohort.

The United States, an early adopter of rotavirus vaccination in 2006, experienced a dramatic decline in the rotavirus disease burden following introduction of 2 live, oral vaccines (Rotarix, GSK Biologicals; RotaTeq, Merck & Co.) [1]. Over the last nearly 2 decades, rotavirus vaccines have been introduced into immunization programs in over 120 countries [2], resulting in a 59% reduction in rotavirus hospitalizations among children <5 years of age globally [3]. Despite substantial progress, natural and vaccine-induced immunity have been consistently shown to be the lowest in high-mortality settings where the rotavirus burden is greatest [4].

A comprehensive understanding of rotavirus natural history and immune response is essential to identifying and mitigating the underlying causes of differential response to infection and vaccination. Examination of rotavirus-specific serum immunoglobulin A (IgA) and immunoglobulin G (IgG), frequently used immunologic indicators, has been foundational in understanding naturally acquired immunity [5, 6] and vaccine effectiveness [7–9] in infants, as well as the factors that influence them [10, 11]. However, few studies outside of clinical trials have longitudinally examined IgA and IgG [5, 6].

The Pediatric Respiratory and Enteric Virus Acquisition and Immunogenesis Longitudinal (PREVAIL) cohort is a US-based birth cohort that followed children to 2 years of age. Integrating host, wild-type infection, vaccination, and serology data, this study provides an opportunity to comprehensively examine rotavirus infection and immune response in the postrotavirus vaccine era. This analysis describes the dynamics of rotavirus-specific IgA and IgG in early childhood, including changes in IgA following rotavirus vaccination and wild-type infection.

Received 11 August 2025; accepted 27 February 2026; published online 3 March 2026

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The Journal of Infectious Diseases®
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https://doi.org/10.1093/infdis/jiag141

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METHODS

Cohort Enrollment and Follow-Up

PREVAIL is a longitudinal birth cohort study in Cincinnati, Ohio. Mother–child pairs were enrolled from March 2017 to July 2018 and followed weekly until the child's second birthday. Full study methods have been previously described [12]. The study protocol was reviewed and approved by the Centers for Disease Control and Prevention, Cincinnati Children's Hospital Medical Center, and the participating hospitals' Institutional Review Boards.

Specimen Collection

Serum was collected from children at birth (cord blood), 6 weeks, 6 months, 12 months, 18 months, and 24 months of age. All samples from 6 weeks to 24 months were tested by enzyme immunoassay (EIA) for rotavirus-specific IgG and IgA. Birth samples were tested for maternally-derived IgG. Stool samples were collected at birth, weekly, and during episodes of acute gastroenteritis (AGE), defined as ≥3 loose or watery stools and/or ≥1 episode of vomiting within 24 hours. Saliva was collected at enrollment for mothers and at 6 weeks and 12 months of age for children (Supplementary Figure 1). Laboratory techniques and definitions are detailed in the Supplementary Material.

Baseline Data and Vaccination Characteristics

Baseline surveys were administered to compile maternal demographic data and weekly surveys collected child illness symptom information. DNA was extracted from saliva samples to determine *FUT2* genotype ("secretor status") of mothers and children (Supplementary Material) [13]. Prior studies have demonstrated that secretor status impacts breastmilk composition and that secretors may be more susceptible to rotavirus infection and more likely to seroconvert following rotavirus vaccination [8, 14].

Rotavirus vaccination information was abstracted from state registries and provider records and categorized by vaccine series and series completeness. Children with ≥2 vaccine doses and with serum collected prior to their first dose (6-week blood draw) and ≥14 days after their second dose (6-month blood draw) were further categorized as vaccine responders or nonresponders. A child was considered a vaccine responder if a ≥ 3-fold increase in rotavirus-specific IgA was observed between the 6-week and 6-month blood draws [15].

Mother–infant pairs were considered adherent to study protocols if they were active in the study for ≥18 months and submitted ≥70% of weekly biological samples.

Statistical Analysis

First, IgA and IgG geometric mean titers (GMT) among adherent children prior to any known wild-type rotavirus infection were examined by child age and stratified by vaccine series to describe overall GMT patterns during the first 2 years of life.

Next, the median change in IgA GMT following rotavirus vaccination was estimated by comparing titers from the 6-week blood draw (prevaccine) to those at 6 and 12 months (postvaccine). Analyses were restricted to children who received ≥2 vaccine doses ≥14 days prior to the postvaccine draw and who had no known wild-type rotavirus infection based on stool testing before the postvaccine draw. A subanalysis examined the change in IgA among vaccine responders.

Similarly, the impact of a wild-type infection on IgA was estimated by comparing the most proximal serum samples prior to and following detection of wild-type rotavirus in stool. To isolate the impact of the wild-type infection on IgA, analysis was limited to adherent children who neither received a rotavirus vaccine dose nor shed a vaccine strain between blood draws. Change in IgA stratified by infection symptom status and association with cycle threshold (Ct) value was also estimated.

IgA and IgG were examined longitudinally to identify potential wild-type infections missed due to limited or ill-timed stool sample collection. A standard 4-fold change in IgA and/or IgG in sequential serum samples was considered indicative of a missed infection unless there was a known potential cause of the increase (known wild-type infection based on stool sample detection, receipt of a rotavirus vaccine dose, or vaccine strain virus shedding between samples). For comparison, a statistical approach using a Classification and Regression Tree (CART) algorithm was employed to identify a threshold of change in IgA and IgG which predicted a wild-type infection (identified via stool) had occurred. Three measures were assessed via the CART analysis: (1) fold change in titer, (2) absolute change in titer, or (3) absolute change in logged titer. Details are described in the Supplementary Material.

Lastly, a generalized estimating equation (GEE) model was fit to assess the unadjusted and adjusted impact of potential predictors on IgA GMT. The outcome was log-transformed IgA with a Poisson error distribution. The model accounted for clustering by child with an exchangeable correlation structure. Predictors were selected a priori based on literature and included child age, maternal and child secretor status, breastfeeding status at the time of the blood draw, childcare received at the time of the blood draw, presence of other children under 5 years of age in the household, number of prior rotavirus vaccine doses, number of prior infections, and baseline IgG. As a subanalysis, the same model was fit to data stratified by child's year of life and by vaccine series. The rationale for stratifying by vaccine series was 2-fold. First, the assay used for quantifying IgA was more sensitive to Rotarix, precluding direct comparison of titers from Rotarix and RotaTeq vaccinees. Second, prior exploration of this cohort [16] identified strong correlation between vaccine series and demographic and socioeconomic factors, resulting in

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potential confounding of vaccine series-specific impacts. All analyses were conducted in R version 4.4.0 [17].

RESULTS

Among the 245 children enrolled in PREVAIL, 242 had ≥1 serum sample and were analyzed (Supplementary Figure 2). The cohort was demographically diverse; half (51.2%) of mothers self-identified as White and half of children (51.6%) were female (Table 1). Slightly more than half (56.2%) of children were publicly insured. Most children (69.0%) and mothers (73.1%) were genotypically secretors. One hundred and one (41.7%) mother–child pairs were considered adherent. In total, 1179 serum samples were tested. Samples were available for 209 (86.4%) children at birth, 213 (88.0%) at 6 weeks, and 166 (68.6%) at 24 months of age (Supplementary Table 1).

Vaccination Status

Most children (81.8%) completed a full rotavirus vaccine series, 10.3% were partially vaccinated, and 7.9% were unvaccinated (Table 1). The median age (weeks) at first, second, and third dose was 9.1 (IQR 8.7–9.9), 18.1 (IQR 17.6–19.7), and 27.0 (IQR 26.4–27.9), respectively, with no notable difference by vaccine series. A total of 90 (37.2%) children were fully vaccinated by 6 months of age, the majority (91.1%) of whom received Rotarix (the third dose of RotaTeq is not recommended for administration until 6 months of age). Among 207 children with 6-month blood draws, 174 (84.1%) had received ≥2 rotavirus vaccine doses prior to the draw, and 96 (46.3%) had completed their series.

IgG and IgA Patterns During Follow-up

A downward trend in IgG GMT was observed among 93 adherent children with serology samples collected prior to any known wild-type infection based on stool samples (Figure 1*A*). Overall IgG GMT was the highest at birth at 292.0 U/mL (95% CI 211.0–404.0). It dropped to nearly undetectable levels among unvaccinated children by 12 months of age (GMT = 9.2 U/mL, 95% CI 9.2–9.2) and remained similarly low at 24 months. Conversely, IgG GMT remained elevated among vaccinated children throughout follow-up. Among Rotarix and RotaTeq vaccinees, 12-month IgG GMT was 61.0 U/mL (95% CI 30.8–121.0) and 71.3 U/mL (95% CI 54.2–93.7), respectively. IgG among Rotarix and RotaTeq vaccinees remained detectable at 24 months of age with GMTs of 55.7 U/mL (95% CI 15.2–204.0) and 32.5 U/mL (95% CI 24.4–43.4).

At 6 weeks of age, few children had detectable IgA levels (Figure 1*B*). Median 6-week IgA GMT was at or near the assay limit of detection (7.5 U/mL) for all 3 vaccination categories. IgA remained below the limit of detection throughout follow-up among unvaccinated children. By 6 months of age, GMT

Table 1. Demographic, Rotavirus Vaccination, Rotavirus Infection, and Serology Information for Infants Enrolled in the PREVAIL Birth Cohort—Cincinnati, Ohio, 2017–2018 (N = 242)

Category	N (%)
Demographics	
Maternal education	
More than high school	129 (53%)
High school or less	113 (47%)
Maternal race[a]	
White	124 (51%)
Black	107 (44%)
Asian or multiracial	11 (5%)
Maternal insurance status	
Publicly insured	136 (56%)
Privately insured	106 (44%)
Child's sex	
Female	125 (52%)
Male	117 (48%)
Child's secretor status	
Secretor	167 (69%)
Nonsecretor	48 (20%)
Unknown	27 (11%)
Mother's secretor status	
Secretor	177 (73%)
Nonsecretor	51 (21%)
Unknown	14 (6%)
Rotavirus vaccination characteristics	
Final vaccination status[b]	
Fully vaccinated	198 (82%)
Partially vaccinated	25 (10%)
Unvaccinated	19 (8%)
Vaccination series	
Rotarix	96 (40%)
RotaTeq	118 (49%)
Mixed/unknown[c]	9 (4%)
Vaccine responder (≥3-fold increase in IgA following ≥2 doses of rotavirus vaccine)[d]	
Yes	105 (43%)
No	36 (15%)
Unvaccinated/not applicable	101 (42%)
Serum specimens available	
Birth (cord blood)	209 (86%)
6-wk	213 (88%)
6-mo	207 (86%)
12-mo	197 (81%)
18-mo	187 (77%)
24-mo	166 (69%)

Abbreviations: IgA, rotavirus-specific immunoglobulin A; IQR, interquartile range; N, number.

[a]More detailed race categories were not available.

[b]A child is considered fully vaccinated if they have received 2 doses of Rotarix, 3 doses of RotaTeq, or 3 doses of a mixed series.

[c]Of those classified as having mixed vaccination series, all 7 were fully vaccinated (ie, received 3 doses of a rotavirus vaccine). Four received 2 doses of RotaTeq and 1 dose of Rotarix. Three received 1 dose of RotaTeq and 2 doses of Rotarix. Two children classified as "other" had unknown vaccine series; one with one dose and the other with 3 doses.

[d]Vaccine response was assessed among children with at least 2 vaccine doses and who had serum collected prior to their first dose (6-wk blood draw) and at least 14 d after their second dose (6-mo blood draw). Of note, RotaTeq is a 3-dose vaccine series. A total of 23 out of 36 (63.9%) nonseroconverters had not yet received their third dose of the rotavirus vaccine series prior to their 6-mo blood draw.

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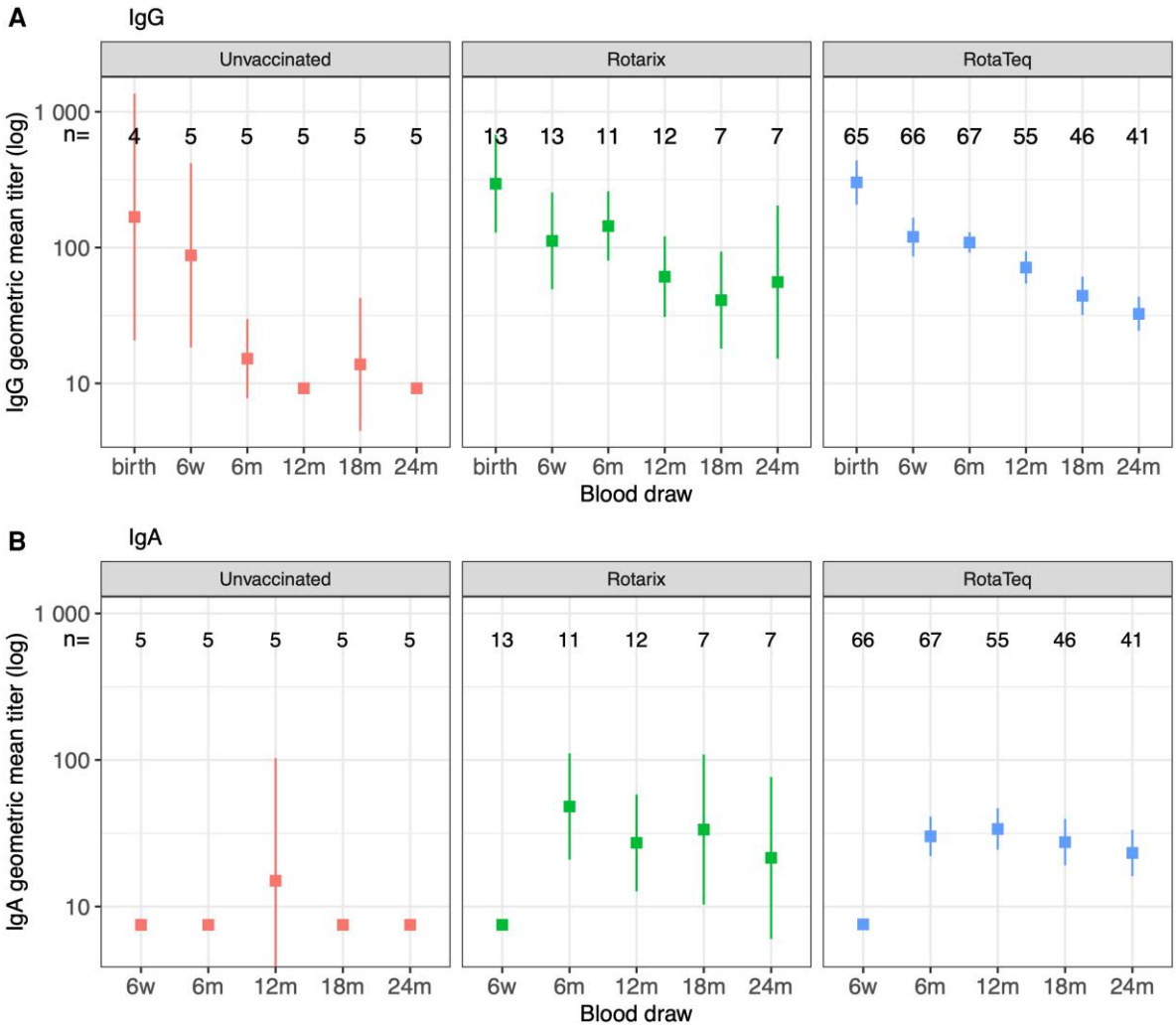

Figure 1. Rotavirus-specific IgA and IgG geometric mean titer and 95% CIs among 93 adherent children in the PREVAIL cohort with no known wild-type rotavirus infection or prior to known wild-type infection based on stool sample testing, stratified by vaccine series and blood draw. Each box represents the geometric mean titer and vertical lines represent the 95% CIs. Includes children with partial or full series vaccination. Two doses of Rotarix or 3 doses of RotaTeq is considered full series vaccination. Abbreviations: CI, confidence interval; IgA, rotavirus-specific immunoglobulin A; IgG, rotavirus-specific immunoglobulin G; m, months; n, number; w, weeks.

rose in vaccinated children to 48.1 U/mL (95% CI 20.9–111.0) and 30.1 U/mL (95% CI 22.1–41.0) for Rotarix and RotaTeq vaccinees, respectively. A 12-month GMT declined to 27.2 U/mL (95% CI 12.7–58.2) for Rotarix and increased slightly to 33.8 U/mL (95% CI 24.4–46.9) for RotaTeq vaccinees. IgA GMT then declined but remained elevated among Rotarix and RotaTeq vaccinated children at 21.5 U/mL (95% CI 6.0–76.6) and 23.2 U/mL (95% CI 16.1–33.3) at 24 months of age.

A subset of 46 (19.0%) children had no detectable IgA throughout the study. These children were demographically similar to those with detectable IgA; however, they had slightly higher IgG GMT at birth ($P = .048$) and were more likely to be unvaccinated ($P < .001$) (Supplementary Table 2). Maternal and infant secretor status was not notably different, though several mothers and children had unknown status.

Rotavirus Vaccine Response and Impact on IgA

Three-quarters (74.5%, n = 105) of children who received ≥2 rotavirus vaccine doses by their 6-month blood draw were considered vaccine responders based on a >3-fold rise in IgA titer (88 [83.8%] of whom also had an IgA titer >20 IU/mL at their 6-month blood draw). Children who completed the full vaccine course by the 6-month draw were more likely to seroconvert ($P = .037$). Child secretor status, baseline IgG titer, and 6-week IgA titer were not associated with seroconversion (Supplementary Table 3).

Assessment of vaccine response at 6 and 12 months was limited to 130 and 115 children, respectively. The median change in IgA GMT from 6 weeks to 6 months was 38.4 U/mL (IQR 4.4–138.1); the change from 6 weeks to 12 months was 23.8 U/mL (IQR 5.8–96.6). The median change in IgA was

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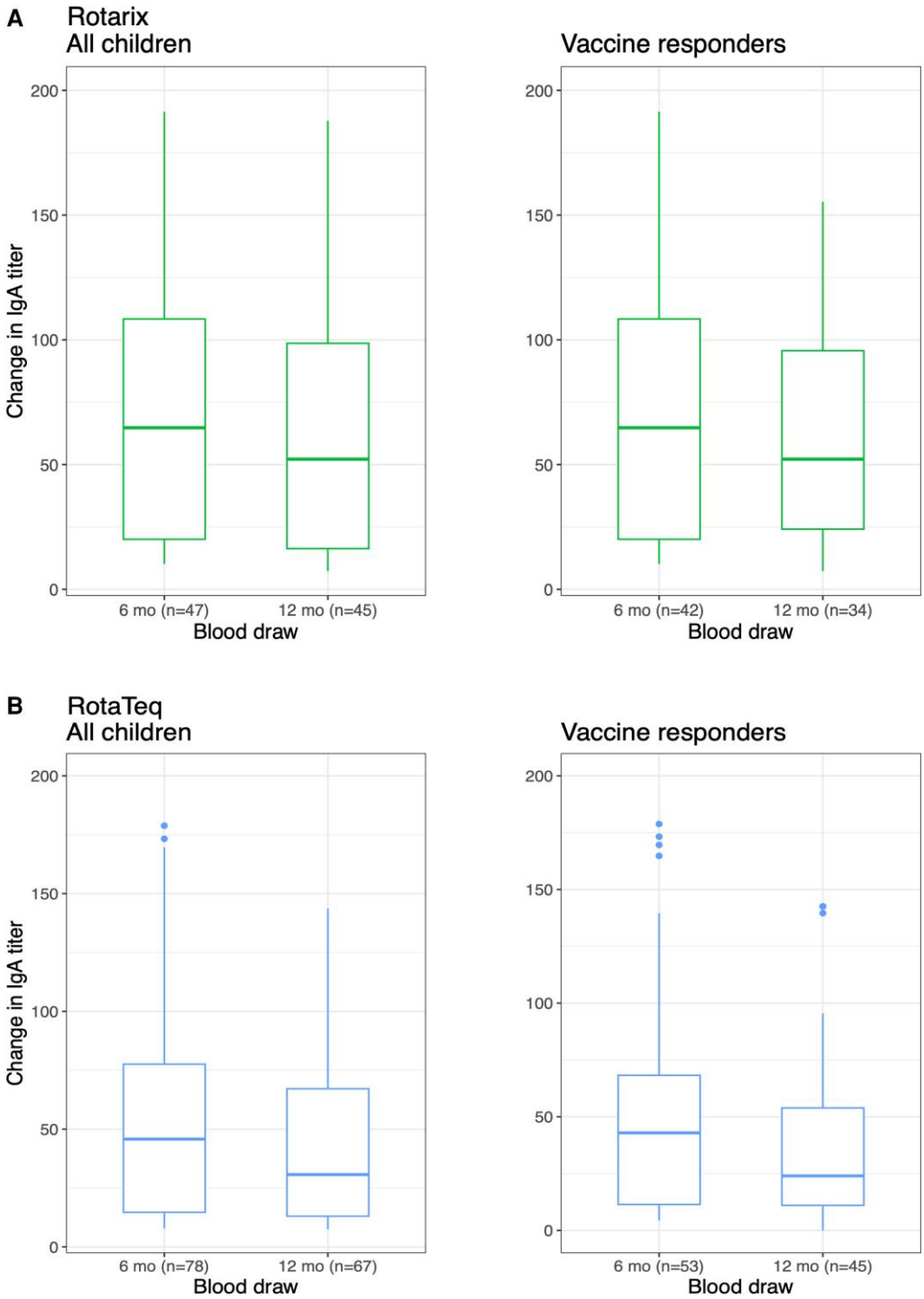

Figure 2. Median change in rotavirus-specific IgA from the 6 week to 6- and 12-month blood draws following receipt of 2 or more doses of the rotavirus vaccine among children in the PREVAIL birth cohort. The middle bar of the boxplot indicates the median change in IgA from the 6-wk blood draw to either the 6-mo or 12-mo blood draw. The upper and lower limits of the box indicate the interquartile range. The whiskers extend to indicate the largest value no further than 1.5 times the interquartile range and dots represent outliers. Rotarix vaccinees who have received 2 doses are considered fully vaccinated. RotaTeq vaccinees require 3 doses to complete the series. Abbreviation: IgA, rotavirus-specific immunoglobulin A; mo, month; n, number.

higher for both timepoints when restricted to vaccine responders (6-month: 64.8 U/mL [IQR 20.3–200.0]; 12-month: 35.8 U/mL [IQR 10.8–101.2]; Figure 2).

Wild-type Rotavirus Infections

The wild-type rotavirus incidence rate was 0.47 (95% CI: 0.39–0.57) infections per child-year with a total of 112 wild-type

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infections observed among the entire cohort. During follow-up, 72 (30.0%) children had ≥1 infection identified via stool with a median age at first infection of 39.8 weeks (IQR 21.6–69.5). Most children (n = 49, 68.1%) completed the rotavirus vaccine series prior to their first infection. Among the 101 adherent children, 44 (43.6%) had ≥1 infection (72 infections total).

Of 98 infections with known symptom status, 22 (22.4%) were symptomatic. The incidence rate of symptomatic infection was 0.09 (95% CI: 0.06–0.14) per child-year. One-quarter of first infections (26.2%, n = 17/65) with known symptom status were symptomatic. One in 4 (23.5%, n = 4/17) symptomatic first infections occurred among children who had not yet received a rotavirus vaccine dose while 58.8% (n = 10/17) had completed the vaccine series prior to infection. Among asymptomatic first infections, 14.6% (n = 7/48) occurred among children who had not yet received a rotavirus vaccine dose while 64.6% (31/48) had completed the vaccine series. The symptom status of first infections did not substantially differ by the number of vaccine doses received prior ($P = .645$). The proportion with symptoms declined by the second infection to 15.4% (n = 4/26 with known symptom status).

Twenty-four adherent children had a single rotavirus infection (27 infections total) identified via stool between 2 serum samples during which time they neither received a rotavirus vaccine dose nor shed vaccine (Table 2). All but 2 infections occurred among secretors. The median time from infection start to subsequent blood draw was 10.6 weeks (IQR 4.1–18.5). The median change in IgA was 64.5 (IQR −2.0 to 192.9) with preinfection and postinfection GMTs of 35.7 U/mL (95% CI: 21.4–59.6) and 102.6 U/mL (95% CI: 41.1–256.0), respectively. All 5 symptomatic infections displayed a substantial increase in IgA, IgG, or both compared to asymptomatic infections, which showed less consistency (Table 2). Symptomatic infections resulted in a median change in IgA of 201.3 (IQR 108.0–2798.2) with preinfection and postinfection GMTs of 65.0 (95% CI: 22.7–186.3) and 727.5 (95% CI: 71.4–7411.0), respectively. Among 19 asymptomatic infections, the median change in IgA was 14.5 (IQR −2.0 to 142.5) with a preinfection GMT of 28.4 (95% CI: 14.9–54.1) and postinfection GMT of 42.8 (95% CI: 19.0–96.2). Ct value was not significantly correlated with absolute change in IgA (rho = −0.31, $P = .198$) or IgA fold change (rho = −0.34, $P = .154$) but was significantly correlated with both measures for IgG (absolute change rho = −0.61, $P = .004$; fold change rho = −0.54, $P = .015$).

Serology-based Identification of Wild-type Rotavirus Infections

Applying the 4-fold change in IgA and/or IgG serologic threshold identified 23 missed rotavirus infections in 22 adherent children (Figure 3). Most infections (n = 20) were identified by meeting the IgG threshold and 3 were identified by IgA. When extended to the entire cohort, 80 infections were identified in 73 children (n = 5, 6.3% by IgA; n = 58, 72.5% by IgG; n = 17, 21.3% by both) (Supplementary Table 5). Combining infections identified by stool and serology, nearly two-thirds (n = 62, 61.4%) of adherent children and half (n = 134, 55.3%) of all children experienced ≥1 wild-type rotavirus infection during follow-up (Supplementary Table 6). Stratifying by total number of infections (stool- and serology-based identification) and blood draw demonstrated a general pattern of increasing IgA GMT with increasing number of previous infections; however, the pattern was not consistent across all blood draws (Figure 4*A*). A similar pattern was observed for IgG GMT (Figure 4*B*).

The training dataset for establishing the CART-based serologic thresholds included 47 children with 13 stool-identified infections available for analysis (Supplementary Table 4, Supplementary Figure 3). The CART-based thresholds were not found to perform better than the standard 4-fold change approach (Supplementary Figures 4–6). When applied to the entire cohort, the CART-based thresholds identified a total of 79, 52, and 43 missed infections for fold change, absolute change, and absolute change in logged IgA and IgG, respectively (Supplementary Table 6).

Multivariable Analysis

In multivariable analyses, the factors associated with IgA level were secretor status, number of prior infections, and vaccination status (Table 3). IgA was higher when the mother and/or child was a secretor compared with when both were nonsecretors (adjusted ratio of means [RoM] = 1.3, 95% CI: 1.1–1.4). This association remained strong in the first (adjusted RoM = 1.2, 95% CI: 1.1–1.4) and second years of life (adjusted RoM = 1.3, 95% CI: 1.1–1.6; Supplementary Table 7). Overall, rotavirus vaccination was strongly associated with IgA, with fully or partially vaccinated children having higher IgA than unvaccinated children. The RoM for fully vaccinated children declined from 1.9 (95% CI: 1.7–2.0) in the first year of life to 1.3 (95% CI: 1.0–1.7) during the second year (Supplementary Table 7). The RoM for vaccination status was similar by vaccine series (Supplementary Table 8). Overall, by year of life and by vaccine series, the number of previous infections was associated with a modest increase in IgA (RoM range: 1.1–1.2). Baseline IgG, location of childcare, and breastfeeding were not associated with changes in IgA, while the presence of children under 5 years of age in the household seemed to have a potential impact only among children who received Rotarix.

DISCUSSION

This study comprehensively examined the dynamics of rotavirus-specific immune response over the first 2 years of life, including the impact of host characteristics, infection, and vaccination. In this birth cohort followed more than a

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Table 2. Characteristics of 27 Wild-type Rotavirus Infections Among 24 Adherent Children in the PREVAIL Cohort With Known Infections Based on Stool Sample Detection of Rotavirus and No Other Known Changes[a] Between Blood Draws

								IgA				IgG Absolute			
Sex	Secretor status	Vacc. doses	Age at infect. (wks)	Time since infect. (wks)	Symptom status[b]	Genotype	Ct value[c]	Prior	Post	Absolute change	Fold change	Prior	Post	change	Fold change
F	Secretor	3	70	7	Asymptomatic	G2P[nt]	39.6	101	61	−40	0.6	261	101	−160.3	0.4
M	Secretor	2	75	5	Unknown	G2P[nt]	21.2	133	98	−35	0.7	132	101	−31.5	0.8
F	Secretor	2	59	18	Asymptomatic	G2P[nt]	39.5	42	8	−34	0.2	207	331	124.5	1.6
M	Secretor	2	56	21	Asymptomatic	G2P[nt]	35.0	29	8	−21	0.3	22	23	0.6	1.0
F	Secretor	3	69	8	Symptomatic	G12P[8]	12.0	147	130	−17	0.9	176	3012	2835.5	17.1
F	Secretor	3	74	4	Asymptomatic	GntP[nt]	37.5	20	13	−7	0.7	201	141	−59.7	0.7
M	Secretor	2	75	2	Asymptomatic	GntP[nt]	39.1	10	8	−3	0.7	9	9	0.0	1.0
M	Nonsecretor	3	42	11	Asymptomatic	GntP[nt]	31.8	8	8	0	1.0	15	68	52.9	4.5
F	Secretor	3	91	14	Asymptomatic	G12P[8]	36.2	8	8	0	1.0	9	9	0.0	1.0
F	Secretor	3	63	14	Asymptomatic	G2P[nt]	38.6	8	16	8	2.1	78	52	−25.8	0.7
F	Secretor	2	54	23	Asymptomatic	Eq-like G3P[8]	34.0	12	23	11	1.9	101	55	−45.7	0.5
M	Secretor	3	69	8	Asymptomatic	G2P[nt]	33.2	8	26	18	3.4	9	140	130.9	15.2
M	Secretor	2	40	12	Asymptomatic	GntP[nt]	37.0	8	29	21	3.8	67	22	−45.4	0.3
M	Secretor	2	81	23	Symptomatic	G12P[8]	21.1	147	255	108	1.7	68	930	862.0	13.6
F	Secretor	3	94	19	Asymptomatic	G3P[8]	33.2	130	251	120	1.9	3012	515	−2496.8	0.2
M	Secretor	3	90	15	Asymptomatic	G12P[nt]	31.0	98	237	139	2.4	47	626	578.7	13.2
F	Secretor	3	71	33	Asymptomatic	G12P[nt]	37.3	8	151	144	20.1	1174	293	−880.9	0.2
M	Secretor	3	54	23	Asymptomatic	G12P[8]	23.4	58	215	157	3.7	51	545	493.4	10.6
M	Secretor	2	101	3	Unknown	G6P[6]	17.1	8	181	174	24.1	9	16	7.1	1.8
F	Secretor	2	77	0	Asymptomatic	GntP[4]	34.5	355	554	199	1.6	2680	2522	−157.9	0.9
M	Secretor	3	96	8	Symptomatic	G12P[8]	28.0	21	223	201	10.4	23	4823	4800.1	207
M	Secretor	2	62	15	Asymptomatic	G12P[8]	19.1	66	311	245	4.7	262	7431	7169.6	28.4
F	Secretor	2	85	20	Asymptomatic	GntP[4]	31.5	156	804	648	5.1	27	738	710.7	27.4
M	Nonsecretor	3	78	1	Symptomatic	G2G4P[4]	29.7	67	2865	2798	42.7	115	7960	7844.8	69.2
M	Secretor	3	57	0	Symptomatic	G12P[8]	14.4	37	9584	9547	258.2	61	443	381.9	7.3
F	Secretor	3	103	1	Unknown	G12P[8]	27.2	143	56 710	56 567	396.2	53	42 663	42 610.1	808.3
M	Secretor	0	4	4	Asymptomatic	G12P[nt]	34.5	NA	8	NA	NA	611	326	−284.3	0.5

Abbreviations: Eq-like, equine-like; F, female; IgA, immunoglobulin A; IgG, immunoglobulin G; Infect, infection; M, male; nt, not typable; Vacc, vaccine.

[a]An infection was excluded if a child received a rotavirus vaccine dose, shed rotavirus vaccine, or had multiple rotavirus infections between blood draws.

[b]A rotavirus infection was considered symptomatic if the first rotavirus-positive stool sample occurred up to 4 d before through 14 d after the start of acute gastroenteritis symptoms. An infection was asymptomatic if the first positive stool was collected more than 4 d before or more than 14 d after symptom start. An infection had unknown symptom status if symptom survey data were missing.

[c]Lowest cycle threshold value identified among stool samples collected during infection.

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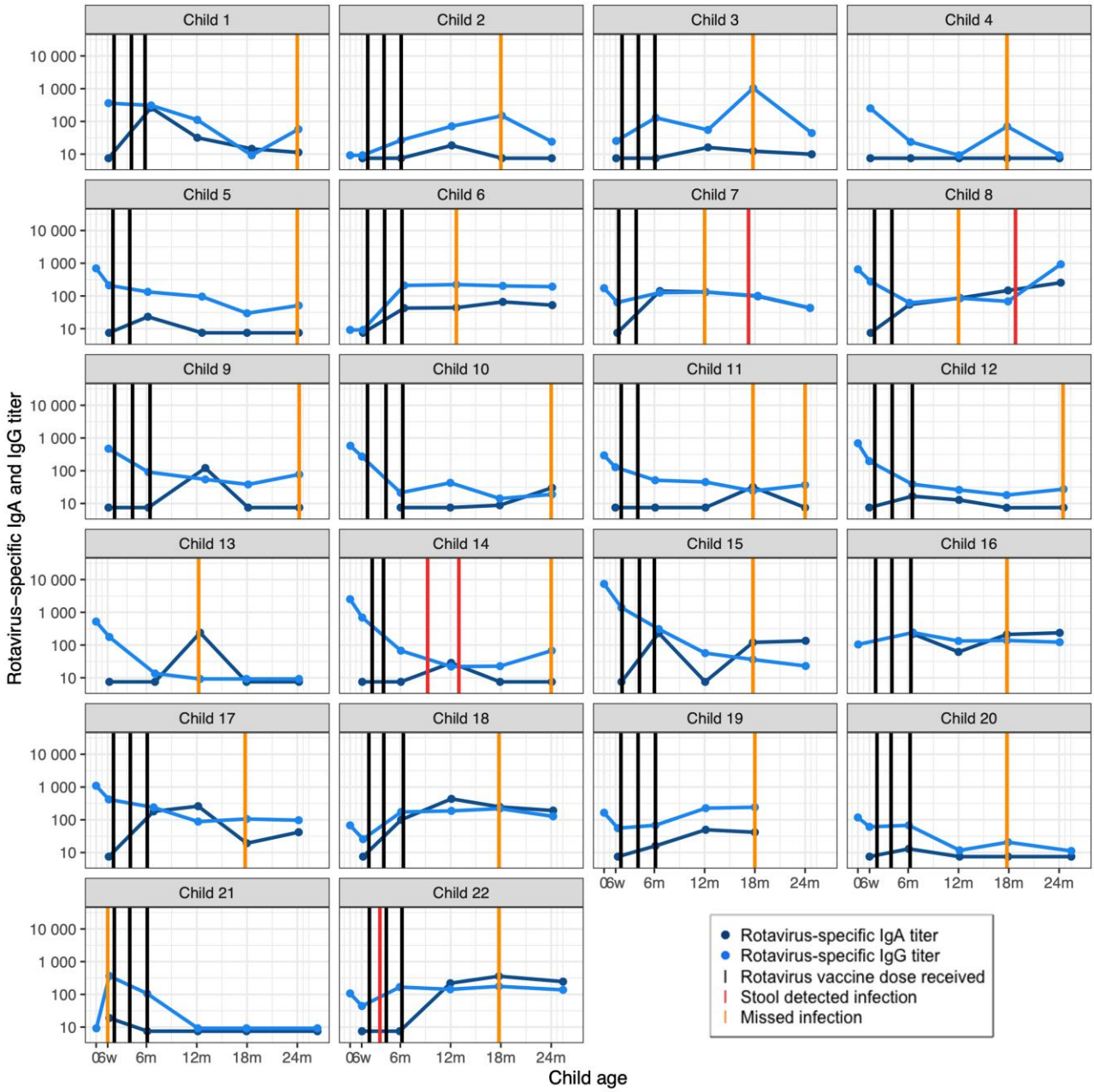

Figure 3. Rotavirus-specific IgA and IgG titer and timing of missed rotavirus infections identified via serology among all adherent children in the PREVAIL cohort with no other known changes between proximal blood draws. The orange line indicates a missed infection identified via serology using a 4-fold change in IgA and/or IgG threshold. The line indicates the time when the increase in IgA and/or IgG was identified. The infection could have occurred anytime between that blood draw and the previous blood draw. Analysis was limited to sequential serum samples between which no other known causes of an increase in IgA or IgG occurred, including receipt of a vaccine dose or vaccine shedding. Abbreviations: IgA, rotavirus-specific immunoglobulin A; IgG, rotavirus-specific immunoglobulin G; m, months; w, weeks.

decade after initiation of the US rotavirus vaccination program, we observed sustained elevation of rotavirus-specific IgA following vaccination. Despite being highly vaccinated, nearly half of adherent children experienced one or more wild-type rotavirus infections by 2 years of age, with symptomatic infections consistently leading to strong immune responses. In adjusted analyses, vaccination status and mother/child secretor status most strongly impacted IgA, though the impact of vaccination substantially decreased in the second year of life.

Rotavirus-specific IgA was elevated among vaccinated children up to 2 years of age, even without the immune boosting effects of wild-type infection during this period. The high rates of seropositivity suggest vaccine-induced immunity remains, albeit reduced, nearly 2 years postvaccination. These results

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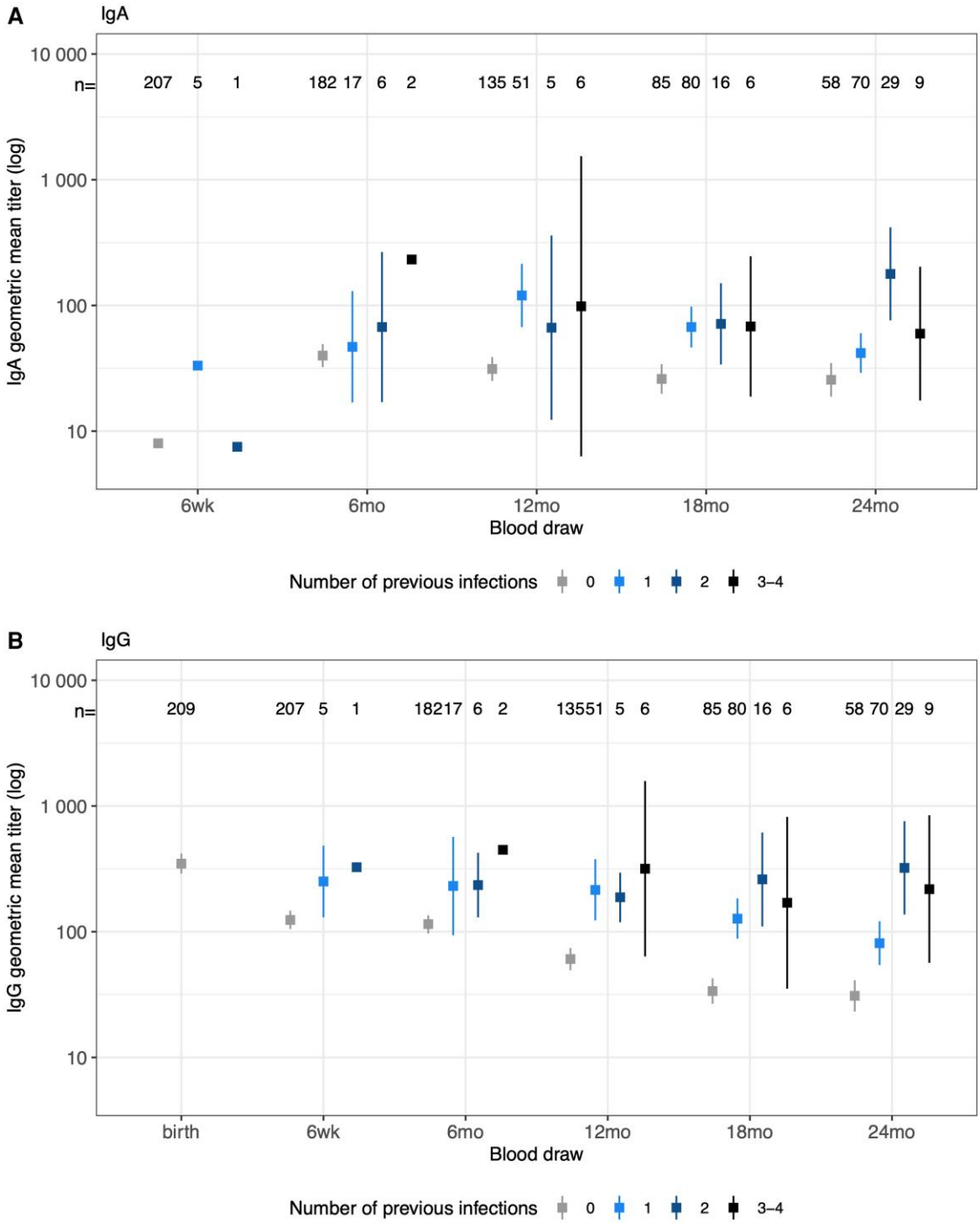

Figure 4. Rotavirus-specific IgA and IgG geometric mean titers and 95% CIs for children in the PREVAIL cohort stratified by blood draw and number of previous rotavirus infections detected by stool and/or serology combined. Missed infections were defined as a 4-fold change in IgA and/or IgG in sequential blood draws with no other known reason for an increased in IgA and IgG (ie, an infection identified via stool, receipt of a vaccine dose, or vaccine shedding. Each box represents the geometric mean titer and vertical lines represent the 95% confidence intervals. Abbreviations: IgA, rotavirus-specific immunoglobulin A; IgG, rotavirus-specific immunoglobulin G; mo, month; n, number of children; wk, week.

align with observations of high rotavirus vaccine effectiveness and relatively well-sustained protection against symptomatic infection provided by vaccination in high-resource settings in comparison to resource-limited and high child mortality settings where both IgA response and vaccine effectiveness are more muted [3, 7]. Detailed longitudinal serologic studies in resource-limited settings, including data on both wild-type infection and vaccination, would allow for comparison of

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Table 3. Unadjusted and Adjusted GEE Model[a] Results Describing the Relationship Between Maternal/Child Characteristics and Child Rotavirus-specific IgA Titer (Log Scale) During the First 2 Years of Life Among Children in the PREVAIL Cohort

Predictor	Unadjusted ratio of means	Adjusted ratio of means
Breastfeeding status		
Exclusively breastfed	1.0 (reference group)	1.0 (reference group)
Partially breastfed	1.0 (0.9–1.2)	1.1 (1.0–1.2)
Never breastfed	1.4 (1.2–1.5)	1.1 (1.0–1.2)
Childcare		
Any care outside home	1.0 (reference group)	1.0 (reference group)
Solely at home	0.8 (0.8–0.9)	1.0 (1.0–1.1)
Other children under 5 y of age in household		
No	1.0 (reference group)	1.0 (reference group)
Yes	1.0 (0.9–1.1)	1.1 (1.0–1.1)
Combined secretor status		
Mom and baby both nonsecretor	1.0 (reference group)	1.0 (refeference group)
Mom and/or baby secretor	1.2 (1.1–1.4)	1.3 (1.1–1.4)
Number of prior infections identified via stool and/or serology[b]	1.3 (1.2–1.3)	1.1 (1.1–1.2)
Vaccination status at time of draw		
Unvaccinated	1.0 (reference group)	1.0 (reference group)
Partially vaccinated	1.7 (1.6–1.8)	1.6 (1.5–1.7)
Fully vaccinated	1.9 (1.8–2.0)	1.8 (1.7–1.9)
IgG titer (log) at birth	1.0 (0.9–1.0)	1.0 (0.9–1.0)

Abbreviations: Adj, adjusted; GEE, generalized estimating equation; IgA, rotavirus-specific immunoglobulin A.

[a]Unadjusted and adjusted estimates were produced by fitting a GEE model with an exchangeable correlation structure and clustering by child.

[b]A serologically identified rotavirus infection was defined as a 4-fold change in IgA and/or IgG in sequential blood draws with no other known reason for an increased in IgA and IgG (ie, a stool identified infection, receipt of a vaccine dose, or vaccine shedding).

results and could provide insight into how durability of IgA and IgG differ by setting.

The change in IgA following natural infection varied widely; however, when aggregated across the cohort, contributed to a pattern of stronger immune response produced by symptomatic infections and increased immunity as the number of infections increased. This pattern provides serologic insight into observed patterns of rotavirus gastroenteritis among young children who acquire increased protection against illness with each infection [5, 6]. The observed individual variability in IgA may be reflective of the range in naturally acquired immunity described globally, with strong immunity and protection against disease attained rapidly in some settings and less protection acquired in others [5, 6, 18, 19].

Results of the multivariable model demonstrate the role of vaccination, particularly in the first year of life, and genetic susceptibility to rotavirus as drivers of immune response in this US setting. Both factors significantly contributed to IgA even after adjusting for other child characteristics. The strong impact of combined maternal/child secretor status was observed in both the first and second year of life, demonstrating the persistence of this relationship in early childhood. The lack of association between IgA and predictors such as baseline IgG and breastfeeding status may reflect their short-term impact on IgA trends. These factors may impact vaccine-induced seroconversion, as shown in prior analyses from this cohort [16] but do not seem influential in predicting longer-term IgA trends estimated in this analysis.

In this cohort, using a 4-fold change in IgA and IgG to identify missed infections led to an increase in the total number of infections by 71%. The traditionally used 4-fold change threshold in IgA and IgG performed equally well or better than the CART-based thresholds in properly identifying known wild-type infections. Measurement and inclusion of an IgG threshold dramatically strengthened our ability to capture missed infections given the short half-life of IgA [20] and emphasizes the importance of both immunologic measures in serological studies.

Several limitations should be noted. First, the immune assay used to measure the IgA and IgG responses mainly measured antibody directed against the VP6 protein of rotavirus. This highly conserved protein is the most abundant protein in the virus and allows measurement of antibody directed against the 2 vaccine types and natural rotaviruses. However, we were unable to directly compare IgA and IgG titers by vaccine series since the assay used a virus more similar to Rotarix than RotaTeq. Results stratified by vaccine series provide encouraging evidence that the patterns observed in these groups were similar. In addition, stool sample collection ended at 24 months of age, though the 24-month blood draw for some children was delayed due to COVID-19 pandemic-related disruptions (Supplementary Figure 7). Rotavirus infections that occurred subsequent to the completion of stool sample collection but prior to the final blood draw could only be identified via serology. Lastly, the use of RT-PCR for detection of rotavirus in stool may overestimate the number of true rotavirus infections when compared with detection via traditional enzyme immunoassay (EIA) methods [21], particularly when coupled with high Ct values (low viral loads). All detected infections were included in this analysis, regardless of symptom status or clinical significance, due to the possibility that peak shedding was missed during weekly stool sampling and, importantly, because this study aimed to assess the relationship between any rotavirus exposure and immune response.

This longitudinal assessment of rotavirus-specific immune response further informs our understanding of rotavirus-specific IgA and IgG and the multifactorial drivers of rotavirus immunity in a postvaccine setting. The associations observed here provide a valuable assessment of immune response in a relatively low transmission setting compared with other settings where complex interactions between

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environmental and host factors continue to challenge and confound analyses. Similar studies conducted across child mortality settings may illuminate key differences in immune response and guide vaccine development and approaches.

Supplementary Data

Supplementary materials are available at *The Journal of Infectious Diseases* online (http://jid.oxfordjournals.org/). Supplementary materials consist of data provided by the author that are published to benefit the reader. The posted materials are not copyedited. The contents of all supplementary data are the sole responsibility of the authors. Questions or messages regarding errors should be addressed to the author.

Notes

Acknowledgments. The authors gratefully acknowledge the participation of the PREVAIL birth cohort families as well as the staff at participating hospitals who assisted with this research.

Author Contributions. M. A. S. and M. M. M. conceptualized this study. J. M. B., C. P. M., R. M. B., and S. C. C. performed data curation. J. M. B. performed formal analysis and data visualization. S. M.-R., M. C. C.-M., and B. W. performed laboratory analyses and results interpretation. A. L. M. and M. A. S. acquired financial support. A. L. M., D. C. P., R. G., U. D. P., J. E. T., M. A. S., and M. M. M. provided oversight. J. M. B. drafted the manuscript. M. A. S. and M. M. M. critically reviewed and edited the manuscript. All authors reviewed the manuscript and had the final responsibility for the decision to submit for publication.

Disclaimer. The findings and conclusions in this report are those of the authors and do not necessarily represent the views of the US Centers for Disease Control and Prevention.

Data availability statement. The data underlying this article cannot be shared publicly for the privacy of individuals that participated in the study. The data may be shared on reasonable request to the corresponding author.

Financial support. This work was funded by a cooperative agreement from the United States Centers for Disease Control and Prevention (IP16-004), NIEHS (T32 ES010957), Open Philanthropy/Good Ventures Foundation and NIAID (U01A1144673). Recipients of the funding were A. L. M. and M. A. S. at Cincinnati Children's Hospital Medical Center, Division of Infectious Diseases. SCC additionally received financial support from MOM2Child R01HD109915.

Potential conflicts of interest. D. C. P. has received institutional awards from Pfizer, Cepheid, and Bavarian-Nordic and was a paid consultant for Merck and Moderna. M. A. S. has received grant funding from Merck, Pfizer and Cepheid and was a paid consultant for Merck. M. M. M. received funding from Merck, GSK and Sanofi. All other authors report no potential conflicts. All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

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